

7 complete genomes (PRJNA357031), RefSeq annotation

33,170 CDS · 9 chromosomes, 2 plasmids

A · GFF text parsing

unique name/product: 1,979

named once in all 7: 1,022

B · Protein homology (mmseqs2)

no homolog in other 6: 7,411

1:1 in all 7 genomes: 979

C · 18-mer uniqueness

unique fraction of the
primary chromosome: 65-92 %

Eligible candidate genes

6,939 genes · route A or B, chromosomal, not pseudo, > 5 kb from mobile elements, >= 300 bp

Sequence-aware filter: unique DNA + single copy

5,087 genes on the dnaA chromosome · >= 150 bp of unique 18-mers, one BLAST hit in own genome

Primer / probe design (primer3)

1,078 assays on 552 genes · 40 core + 40 unique genes per strain, 70-150 bp amplicons

Exhaustive in-silico PCR (bowtie1, <= 3 mismatches, 7 genomes + plasmids)

716 assays pass · one on-target product, no near-perfect off-target primer site

Multiplex selection

42 assays (top 6 genes per strain) -> 279,936 7-plex combinations scored on dimer dG

Final 7-plex, verified in silico

pooled PCR gives exactly 7 products · knock-out test 7/7 · worst between-assay dG -5.64 kcal/mol